A Multi-Expert Agentic Pipeline for Peptide Generation

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Abstract

We present a proof-of-concept multi-expert agentic pipeline for *in silico* peptide generation [1]. A central orchestrator coordinates three specialized agents: an AI Designer that proposes candidates, a Chemistry Constraints agent that enforces proxy feasibility rules (length, net charge, hydrophobicity, and forbidden motifs), and a Biology Function/Risk agent that critiques expected activity and toxicity-related risks [2].

The pipeline iterates through a propose → filter → score → revise loop, with round-level logs for traceability. Evaluation is fully quantitative and reports validity, average and Top-K score, diversity, and constraint pass rates against a non-agentic baseline. Results show that iterative multi-agent coordination improves both constraint satisfaction and top-ranked candidate quality over single-pass generation [1–3].

Keywords: Multi-Agent Systems, Peptide Generation, In Silico Drug Design, Variational Autoencoder, Conditional Variational Autoencoder, Evolutionary Scale Modeling.

1 Introduction

Peptides, and especially antimicrobial peptides (AMPs), are promising candidates for therapeutic, agricultural, and industrial applications [1]. Their *in silico* design is typically formulated as an iterative optimization process: generate candidate sequences, apply formal constraints, and evaluate each candidate with proxy scores for activity, toxicity risk, and stability [1, 2]. While recent machine learning methods have accelerated sequence design, balancing multiple and sometimes conflicting objectives remains difficult in practice.

This project addresses that challenge through an agentic proof of concept in which specialized software agents cooperate in a closed optimization loop. The pipeline includes an AI Designer agent that proposes candidates, a Chemistry Constraints agent that filters feasibility using proxy physicochemical rules, and a Biology Function/Risk agent that critiques expected function and risk. An orchestrator manages the iterative cycle (propose $\rightarrow$ filter $\rightarrow$ score $\rightarrow$ revise), applies stopping criteria, and stores logs for traceability.

The work is centered on AI aspects (modeling, orchestration, reasoning, and quantitative evaluation) and assumes no prior expertise in wet-lab biology or chemistry. Evaluation is explicit and reproducible: we measure validity, average and Top-K score, diversity, pass rates for constraints, and score evolution across rounds, then compare against a non-agentic baseline. This framing allows us to test whether multi-expert coordination provides measurable gains over single-pass generation .

The remainder of this paper is structured as follows: Section 2 provides a brief overview of the project context and related work. Section 3 details the architecture of our multi-agent pipeline and the specific roles of the AI, Chemistry, and Biology agents. Section 4 outlines our experimental setup and quantitative evaluation metrics. Section 5 presents our findings, including baseline comparisons and ablation studies. Finally, Section 7 concludes the report and discusses potential future work.

2 Context and Related Work

TODO: Discuss existing methods for peptide generation (traditional machine learning vs. generative AI). TODO: Introduce the concept of Agentic Workflows / Multi-Agent LLM systems. TODO: Define the scope (AI orchestration focus, no deep biology/chemistry background assumed, solely in silico with proxy scores)[cite: 7, 38, 39].

3 Multi-Agent System Architecture

The global pipeline is organized around three specialized agents coordinated by a central orchestrator. Figure 1 summarizes the end-to-end data flow and feedback loops in the iterative process.

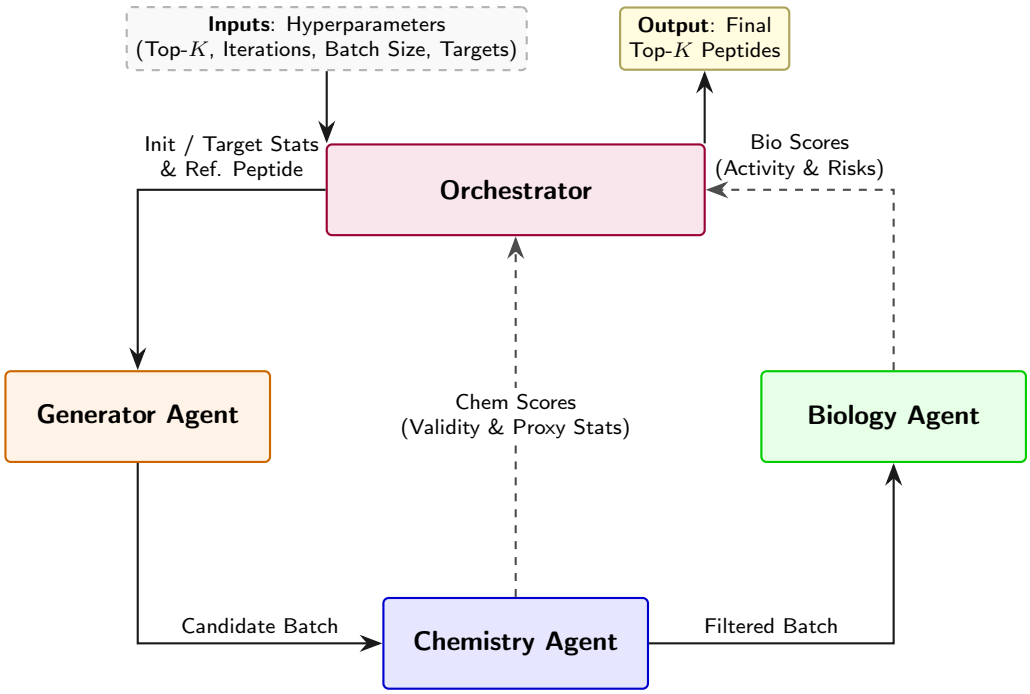

Figure 1: Agentic Pipeline Architecture. Solid lines denote the sequential flow of peptide candidate batches. Dashed lines indicate the feedback loops, where both the Chemistry and Biology agents return their respective proxy scores back to the Orchestrator. Color coding differentiates the specialized domains of the pipeline.

3.1 The Orchestrator

The Orchestrator is the control layer that turns the three experts into a single iterative optimization system. Its role is to enforce a fixed sequence of operations:

1. Request a batch of candidates from the Generator (if not given when originally called)
2. Pass this batch through the Chemistry Agent to keep only chemically admissible peptides and associated proxy descriptors, and send the surviving candidates to the Biology Agent for biological scoring.
3. Based on these expert outputs, the Orchestrator ranks candidates, keeps the Top-K subset, and uses this subset as the seed for the next generation cycle.

In this way, each iteration combines exploration (new proposals) and exploitation (retaining best current candidates) under one consistent decision policy. The stopping criterias are handled at the orchestration level rather than inside each expert module.
In practice, the loop can stop when :

- A maximum number of iterations is reached
- No valid candidates survive the chemistry gate

- The optimization objective is considered stable enough according to user-configured run settings (for example fixed Top-K, batch size, and iteration budget)

This implementation allows reproducible exepimentations : changing run behavior only requires updating orchestrator-level parameters, there is no need to change the experts. This agent also provides traceability by recording round-wise pipeline events and outcomes:

- Input settings
- Filtered populations
- Expert scores
- Selected candidates

This log-centric design makes the optimization path auditable from end to end: one can reconstruct why a peptide was retained or discarded at a given step, compare runs under different settings, and support post-hoc analysis of convergence behavior and failure cases.

3.2 Generator Agent

TODO: Describe how this agent generates candidates (prompting strategies, templates, mutation/crossover operators)[cite: 15].

3.3 Chemistry Agent

The Chemistry Agent is the deterministic constraint-and-scoring stage of the pipeline. At a global level, it receives peptide sequences from the generator, evaluates them against user-defined physicochemical targets, and returns both a feasibility decision and a quantitative score used downstream by the orchestrator. The behavior of this agent is driven by a configurable target profile (ranges, target values, optional weights, and pH), which allows the same workflow to be reused under different experimental objectives.

Its workflow can be summarized through three complementary functions:
1. Sanitization : Generative models can hallucinate sequences that are syntactically correct but chemically invalid or unstable. The Chemistry Agent goes beyond simple string matching by attempting to instantiate a molecular graph for each sequence using the RDKit framework. This process involves a full sanitization step: checking valence rules, aromaticity, and atom consistency. Any sequence that fails to convert into a valid RDKit Mol object is immediately discarded as a chemical syntax error. This ensures that downstream agents only process realizable matter.

2. Multidimensional Property Profiling : To characterize developability, the agent computes a configurable set of descriptors from two complementary

sources: peptide-centric properties (length, net charge at chosen pH, isoelectric point, hydrophobicity, cationicity) and molecule-centric properties (notably RDKit-based $\log P$). This mixed profile lets the orchestrator compare candidates in a unified physicochemical space.

3. Quantitative Scoring and Adaptive Filtering : The agent assigns each candidate a continuous score based on distance to target values, with optional per-property weighting. It then applies a two-step filtering policy: first keep peptides that satisfy all hard limits, then (if too few survive) refill the population with near-miss candidates that are closest to the target profile. This adaptive mechanism prevents early pipeline collapse while preserving optimization pressure.

The Chemist agent is the first called agent after the generator is used. It allows a first selection and analysis on the generated peptides list before sending them to the Biologist agent. Based on the strict limits provided by the user it sends back to the orchestrator scores and properties on the peptides. From an higher-level perspective, the returned data acts as a deterministic generated ground-truth wich can be trusted by the Biologist agent for its prediction.

3.4 Biology Agent (Function & Risk)

The Biology Agent evaluates the generated candidates and provides the biological signal used by the iterative loop. In our implementation, it is based on Meta/Facebook's Evolutionary Scale Modeling family (ESM-2) [3,4] and can be conditioned by context (for example an existing peptide reference or requested target properties).

At each iteration, the Biology Agent scores the current valid batch and returns values used by the orchestrator to rank candidates and keep the Top-K peptides for the next round. This makes it the core selection component on the biological side of the pipeline.

As a design choice, this score is treated as a relative proxy to guide search, not as a direct experimental measurement. The detailed scoring formulation is reported in Section 4.1.

4 Experimental Setup

TODO: Describe the technical implementation (e.g., framework used, LLM endpoints).

4.1 Quantitative Scoring System

Biologist Agent Scoring

The objective is to rank a batch of peptides and identify which candidate is closest to the provided context (an existing peptide or requested properties represented as a reference embedding).

The implementation uses ESM-2 embeddings (default: `facebook/esm2_t12_35M_UR50D`) with attention-mask-aware mean pooling to produce one vector per peptide. We first tested cosine similarity between candidate and reference embeddings. In practice, cosine values were often too similar for short peptides in this protein-wide latent space, which reduced ranking contrast. We therefore switched to Euclidean distance, then normalized batch distances with a Z-score and mapped them to $[0, 1]$ with a sigmoid.

The resulting score is used for relative ranking inside each batch and for Top-K selection by the orchestrator. It should be interpreted as a proxy signal to guide the search process, not as an absolute biological activity measurement.

Chemist Agent Scoring

Crucially, to prevent pipeline starvation (where strict filters might discard the entire population early in the optimization process), the agent employs an adaptive thresholding mechanism. The workflow operates in two phases:

1. **Hard Constraint Check:** Candidates are first filtered against strict limits (e.g., range constraints on length or charge).
2. **Population Rescue:** If the survival rate falls below a predefined threshold (e.g., 50%), the agent relaxes the hard constraints. It then fills the remaining quota by selecting the top-ranking rejected candidates—those that, while slightly violating limits, are mathematically closest to the ideal TPP.

This strategy ensures a continuous flow of candidates to the Biology Agent while maximizing selective pressure towards developable molecules.

- **Solubility & Stability:** We calculate the Isoelectric Point (pI) and Net Charge at specific physiological pH levels (e.g., pH 5.5). These metrics are critical for predicting aggregation propensity and solubility in biological fluids.
- **Bioavailability & Permeability:** The generated profile includes the partition coefficient ($\log P$), Topological Polar Surface Area (TPSA), and hydrophobicity indices. These values act as proxies for the peptide's ability to cross biological membranes and interact with hydrophobic pockets.
- **Structural Indices:** Metrics such as the Boman Index are computed to estimate potential protein-binding potential or antimicrobial activity.

Orchestrator-Level Scoring and Ranking

TODO: Describe how the orchestrator combines outputs from expert agents (Biologist + Chemist), applies weighting or filtering rules, and computes the final ranking used for Top-K selection.

4.2 Evaluation Metrics

TODO: Define the metrics you tracked: Validity (%), Diversity (distances/entropy/n-grams), and Constraint pass rates[cite: 25, 27, 28].

4.3 Baselines

The baseline is designed as a simpler non-agentic reference to test whether the proposed multi-expert iterative pipeline provides measurable gains. Concretely, we use a standalone Conditional Variational Autoencoder (CVAE) that directly generates peptide candidates from conditioning features, without chemistry/bio critique loops and without orchestrated revision rounds.

This baseline CVAE is trained on the dataset built with `database/get_data.py`. The extraction script queries DBAASP asynchronously and retains sequence-level and physicochemical fields used for conditioning (length, pH, molecular weight, logP, net charge, isoelectric point, hydrophobicity, and cationicity). In practice, many API entries contain missing values for one or more properties. The baseline data pipeline therefore substitutes absent fields with fixed defaults before standardization. This dataset is thus not optimal from a data-quality perspective, but for a proof-of-concept objective we considered it sufficient to establish a consistent comparison point.

From an implementation perspective, the baseline model is a GRU-based CVAE. The encoder embeds amino-acid tokens and maps sequence-condition pairs to latent Gaussian parameters; the decoder is autoregressive and conditioned on both latent code and property vector. Training uses cross-entropy reconstruction loss with KL regularization, Adam optimization, and a progressive KL weight (annealed β).

At inference time, candidate peptides are sampled from the latent prior under a target condition vector (scaled with the training `StandardScaler`), with temperature and top-k sampling to control exploration. This baseline serves only as the comparison anchor: unlike the full pipeline, it performs one-shot conditional generation and does not benefit from multi-agent feedback, constraint-aware iteration, or round-level refinement.

5 Results and Analysis

5.1 Overall Performance and Top-K Gain

TODO: Present the final Top-K peptides. TODO: Show the measurable gain in average score and Top-K score compared to the non-agentic baseline[cite: 13, 26, 30]. Use a table.

5.2 Pipeline vs. Baseline Comparison

TODO: Add a direct comparison table between the full multi-agent pipeline and the standalone CVAE baseline (one-shot generation), using the same eval-

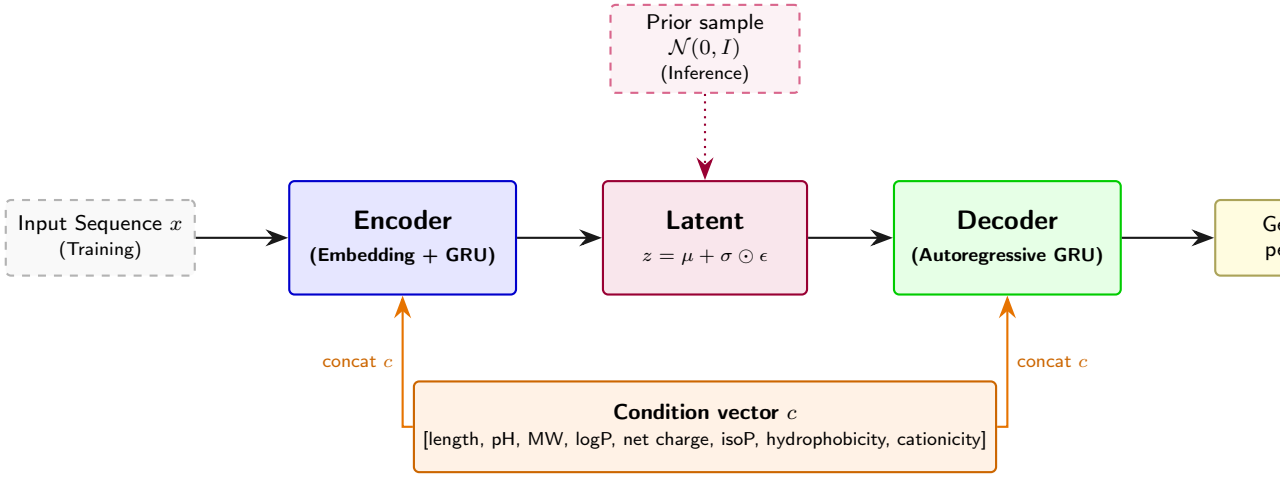

Figure 2: Baseline CVAE architecture used for comparison. During training, the model learns $q_\phi(z \mid x, c)$ and reconstructs sequences with an autoregressive decoder conditioned on z and c. During inference, peptides are generated by sampling $z \sim \mathcal{N}(0, I)$ and decoding once, without iterative multi-agent feedback.

uation protocol. TODO: Report at least validity, average score, Top-K score, diversity, and chemistry-constraint pass rate for both systems. TODO: Add a short interpretation of the observed gap and indicate whether improvements are consistent across runs/seeds. TODO: Explicitly connect this comparison to the baseline definition in Section 4.3 and Figure 2.

5.3 Trajectory of Improvement

TODO: Analyze the evolution of the scores across multiple iterative rounds. Include a graph plotting score vs. iteration number[cite: 29].

5.4 Ablation Studies

TODO: Present the results of running the pipeline with missing components to prove their worth. TODO: Compare the full pipeline vs. "Without Chemistry Agent", vs. "Without Biology Agent", vs. "Without Iteration"[cite: 42].

6 Discussion & Future Work

6.1 Chemist agent improvement

Trough the whole filtering and scoring process by the agent, the computed scores are obtained using a peptide sequence code (e.g "ACDEF", "AA"). The SMILES format is also very popular as its offer better topological information

on the peptide, therefore we could be interested in gathering data from it, in addition to the current format. Unfortunately, our current lack of knowledge on the subject doesn't allow us to be certain about how this additional data could impact our scoring system and the overall pipeline.

6.2 Future Work: Peptide-Specific Biological Scoring

The main improvement is to replace the current proxy with a model trained (or fine-tuned) specifically on peptides. Ideally, this model should score or classify a peptide conditionally on a target (or target-related context), instead of relying on a general protein-space similarity signal.

The expected benefit is a more accurate and reliable biological signal for ranking candidates in the loop. A peptide-specialized model could also provide richer outputs than a single proxy score, for example clearer estimates of expected activity and additional information about risk or toxicity.

In short, our current biologist agent is a practical baseline for iterative optimization, while a target-conditioned peptide model is the natural next step to improve biological relevance and decision quality.

7 Conclusion

TODO: Summarize that the multi-expert agentic pipeline successfully demonstrates iterative improvement in peptide design. TODO: Mention that all code, scripts, and logs are provided in the Git repository[cite: 32, 33].

Bibliography

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[4] Facebook Research, "ESM: Evolutionary Scale Modeling," GitHub repository. [Online]. Available: `https://github.com/facebookresearch/esm`